Thrift Tech Inventory & Sales System (TTISS)

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Client: Thrift Tech – Braamfontein

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Appendix A – Technology Stack Summary

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1. Introduction

To the management of Thrift Tech,

The Thrift Tech Inventory and Sales System (TTISS) is a web-based information system designed to replace the manual, paper-driven processes currently used at Thrift Tech's second-hand clothing retail outlets in Braamfontein. As stated in the Project Plan (Thrift Tech Project Plan Document 2026), the current manual approach leads to inventory inaccuracies, slow customer checkout (with item lookup taking far longer than 10 seconds), lack of sales visibility, pricing inconsistencies, and limited scalability for future online expansion.

The TTISS automates the following core business operations:

- **Inventory management:** Staff and users can add, edit, search, and view clothing items. User-submitted items require manager approval before going live.
- **Sales processing:** Cashiers and customers can add items to a shopping cart, review quantities, and complete checkout, which automatically deducts stock and generates a receipt.
- **Reporting:** Managers can generate weekly sales reports showing total revenue, units sold, top-selling categories, and low-performing items.
- **Monitoring:** The system automatically displays low-stock alerts when inventory drops below configurable thresholds.
- **User access control:** Users log in with email/name and password. Roles include Customer/Staff, Manager, and Administrator.

Target users (from the Requirement Analysis Document, Section 2.1):

- Store Staff / Cashier / Customer
- Store Manager
- System Administrator

Technical environment: The system runs as a web application on existing PC and tablet devices at Thrift Tech, using a standard web browser. The backend uses a three-tier architecture with a MySQL database.

Project objectives from the Project Plan (Task 1):

- 90% reduction in manual inventory errors
- Item lookup time during sales ≤ 10 seconds
- Delivery of weekly sales reports and low-stock reports

The TTISS design presented in this document directly addresses these objectives through its GUI screens (product listing, search, cart, admin dashboard) and database structure.

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2. Logical Architectural Design

2.1 High-Level Architectural Design

Architecture type: Three-tier (client–server) architecture.

Tier	Name	Components	Responsibilities
Tier 1	Client (Presentation)	Web browser (Chrome, Edge, Firefox) on existing Thrift Tech PC/tablet	Renders HTML/CSS/JS pages from Figma-inspired design; captures user input from Login, Signup, Sell, Cart, and Home screens; displays output and reports. No business logic.

Tier	Name	Components	Responsibilities
Tier 2	Application (Business Logic)	Web server (Apache/Nginx) + PHP or Python Flask application	Hosts TTISS application. Contains all functional logic: user authentication, inventory CRUD with approval workflow, shopping cart management, sales processing, report generation, low-stock monitoring.
Tier 3	Data	MySQL database server	Stores all persistent data: inventory items (active and pending), sale transactions, sale line items, user accounts, stock threshold configurations.

Position of the database: The MySQL database resides on the **data tier**, physically or logically separate from the application server. The application tier connects to the database via SQL queries; the client tier never connects directly to the database.

Functional building blocks – allocation to components (mapped to GUI screens):

Function	GUI Screen(s)	Allocated to	Explanation
Display login, signup, home, cart, sell, admin screens	All six screens	Client tier (browser)	HTML/CSS/JavaScript handles presentation only.
Validate login credentials	Login Page	Application tier	Server compares email/name and password hash against USER_ACCOUNT table.

Function	GUI Screen(s)	Allocated to	Explanation
Register new user	Sign-Up Page	Application tier + Data tier	Server validates unique email, inserts new USER_ACCOUNT.
Submit new item for approval	Sell / Product Submission Page	Application tier + Data tier	Server creates INVENTORY_ITEM with is_active = FALSE (pending).
Search and filter inventory	Home / Product Listing Page (search bar, category filters)	Application tier + Data tier	Server builds parameterised SQL query, returns matching items.
Add to cart, update cart, remove item	Home Page + Shopping Cart Page	Application tier (session)	Server maintains cart session; recalculates totals.
Process checkout (sale)	Shopping Cart Page → Checkout	Application tier + Data tier	Server creates SALE_TRANSACTION, SALE_ITEM records, decrements INVENTORY_ITEM.quantity, generates receipt.
Approve/reject pending items	Admin / Manager Dashboard (Approve/Reject buttons)	Application tier + Data tier	Server updates INVENTORY_ITEM.is_active or deletes record.

Function	GUI Screen(s)	Allocated to	Explanation
Remove live product	Admin Dashboard (Remove Product button)	Application tier + Data tier	Server soft-deletes (is_active = FALSE).
Generate sales report	Admin Dashboard (Generate Report button)	Application tier + Data tier	Server aggregates SALE_TRANSACTION + SALE_ITEM data, formats as HTML/PDF.
View low-stock alerts	Admin Dashboard (automatic panel)	Application tier	Server compares INVENTORY_ITEM.quantity with STOCK_THRESHOLD.min_quantity.
Logout	Dashboard (Logout button)	Application tier	Server destroys session.

Referencing previous documents (Requirement Analysis – TASK 2 PDF):

- **Input specifications (Section 3.1):** Login credentials (Login Page), item details (Sell Page), search query (Home Page search bar), sale transaction data (Cart → Checkout), report date range (Admin Dashboard), user account details (Sign-Up Page) – all captured on client tier, processed in application tier.
- **Output specifications (Section 3.2):** Product listing (Home Page), sale receipt (post-checkout), weekly sales report (Admin Dashboard output), low-stock alert (Admin Dashboard panel), confirmation/error messages – generated in application tier, rendered on client tier.
- **Functional requirements table (Section 1.3):**

- FR-01 (Inventory Management) → Sell Page + Admin Dashboard (approval) + Application + Data tiers
- FR-02 (Item Search & Lookup ≤10 sec) → Home Page search + Application tier with indexed queries
- FR-03 (Sales Processing) → Shopping Cart → Checkout + Application + Data tiers
- FR-04 (Sales Reporting) → Admin Dashboard (Generate Report) + Application + Data tiers
- FR-05 (Low-Stock Alerts) → Admin Dashboard alert panel + Application tier
- FR-06 (Pricing Management) → Sell Page + INVENTORY_ITEM table
- FR-07 (User Authentication) → Login Page + Sign-Up Page + USER_ACCOUNT table
- FR-08 (Customer Records) → Can be added later via new CUSTOMER table

2.2 Low-Level Architectural Design

The low-level design shows the relationship between **actors**, **functions (use cases)**, and **database tables**. Each use case is linked to the database table(s) it reads or writes.

Actors (from Requirement Analysis Document, Section 2.1, aligned with GUI roles):

Actor	**GUI Screen(s) Used**	**Role**
Customer / Store Staff	Home Page, Shopping Cart, Sign-Up, Login, Sell Page	Browses products, adds to cart, purchases, submits items for approval
Store Manager	Admin Dashboard, Login	Approves/rejects items, removes products, generates reports, views low stock

Actor	GUI Screen(s) Used	Role
System Administrator	Admin Dashboard (backend)	User account management, database backups, deployment

Functions (use cases) – mapped to GUI screens:

UC-ID	Use Case	GUI Screen	Description
UC-01	Login	Login Page	Authenticate user credentials against USER_ACCOUNT
UC-02	Register	Sign-Up Page	Create new user account
UC-03	Submit Item for Approval	Sell Page	Insert new INVENTORY_ITEM with is_active = FALSE
UC-04	Search Inventory	Home Page (search bar + filters)	Query INVENTORY_ITEM WHERE is_active = TRUE
UC-05	Add to Cart	Home Page (Add to Cart button)	Add item to session cart

UC-ID	Use Case	GUI Screen	Description
UC-06	Manage Cart	Shopping Cart Page	Update quantities, remove items
UC-07	Process Sale (Checkout)	Shopping Cart Page → Checkout	Create SALE_TRANSACTION, SALE_ITEM, decrement stock
UC-08	Approve/Reject Item	Admin Dashboard	Update INVENTORY_ITEM.is_active
UC-09	Remove Product	Admin Dashboard	Soft delete INVENTORY_ITEM
UC-10	Generate Sales Report	Admin Dashboard	Aggregate SALE_TRANSACTION + SALE_ITEM
UC-11	View Low-Stock Alerts	Admin Dashboard	Compare quantity vs STOCK_THRESHOLD
UC-12	Logout	Dashboard (Logout button)	Destroy session

Low-level diagram (actors → use cases → database tables):

Actors	Use Cases	Database Tables
Customer / Staff	UC-01 Login, UC-02 Register, UC-03 Submit Item, UC-04 Search, UC-05 Add to Cart, UC-06 Manage Cart, UC-07 Process Sale	USER_ACCOUNT, INVENTORY_ITEM, SALE_TRANSACTION, SALE_ITEM
Store Manager	UC-01 Login, UC-04 Search, UC-08 Approve/Reject, UC-09 Remove Product, UC-10 Generate Report, UC-11 View Low Stock, UC-12 Logout	USER_ACCOUNT, INVENTORY_ITEM, SALE_TRANSACTION, SALE_ITEM, STOCK_THRESHOLD
System Administrator	UC-01 Login	USER_ACCOUNT

Relationships between use cases (arrows):

- UC-07 Process Sale **includes** UC-04 Search Inventory and UC-06 Manage Cart
- UC-08 Approve/Reject Item **depends on** UC-03 Submit Item for Approval
- UC-10 Generate Sales Report **depends on** data created by UC-07 Process Sale
- UC-11 View Low-Stock Alerts **uses** data from UC-03

Circle notation (database table indicators):

- UC-07 Process Sale ○ SALE_TRANSACTION ○ SALE_ITEM ○ INVENTORY_ITEM

- UC-10 Generate Sales Report ○ SALE_TRANSACTION ○ SALE_ITEM
- UC-11 View Low-Stock Alerts ○ INVENTORY_ITEM ○ STOCK_THRESHOLD

Note comment: INVENTORY_ITEM ○ – Stores all clothing items. Inserted on submission (UC-03), updated on approval/rejection (UC-08) or removal (UC-09), and decremented on sale (UC-07). Queried for search (UC-04) and low-stock alerts (UC-11).

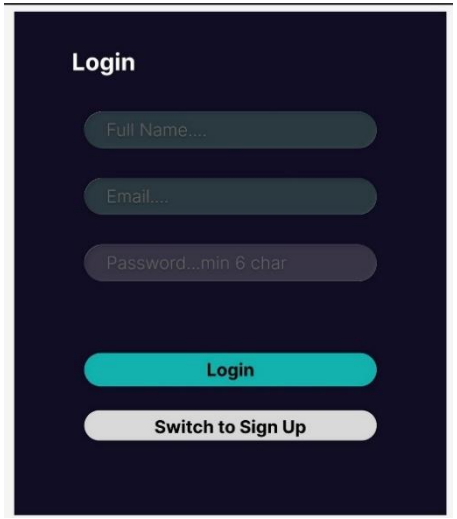
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3. User Interaction Design – Input Interactions (Option 2: Complete GUI Definition)

The TTISS uses a complete graphical user interface (GUI) designed in Figma. The following screens define all input interactions where users enter data into the system.

Figma Prototype Link: https://www.figma.com/design/vq5z23Yxmm94fTEwkLHr6q/thrift_tech_store?node-id=0-1&t=JeE2JPcH6ouuNxe6-1

3.1 Login Page (Input Interaction #1)

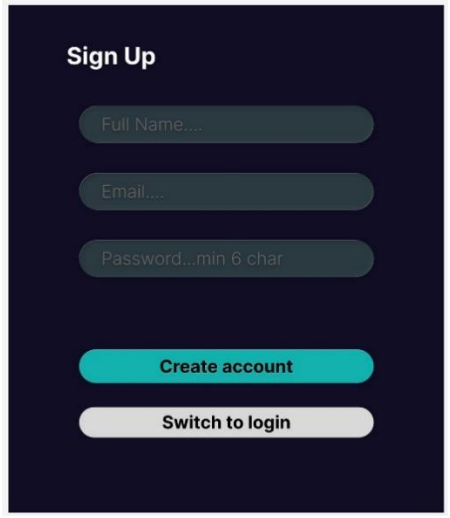


Screen Name: Login Page
Purpose: Authenticates existing users. Verifies credentials to grant access to user-specific features such as selling, cart management, and order history.

Control	Type	Validation	Maps to Database Field
Full Name	Text field	Required, must match existing user	USER_ACCOUNT.full_name
Email	Text field	Required, valid email format, must exist	USER_ACCOUNT.username
Password	Password field	Required, must match hash	USER_ACCOUNT.password_hash
Login Button	Submit button	Triggers authentication	–

Referenced in GUI: *See Figma design – Login Page*

3.2 Sign-Up Page (Input Interaction #2)

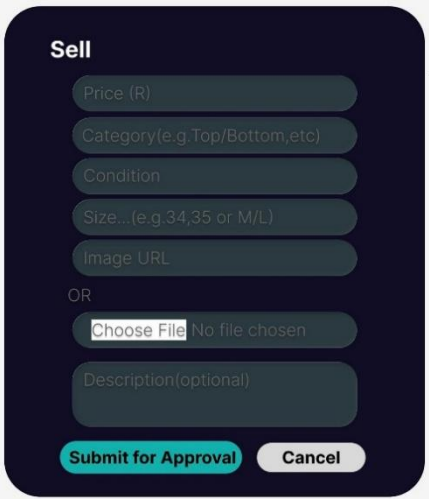


Screen Name: Sign-Up Page
Purpose: Registers new users. Collects necessary information to create a new account on the platform.

Control	Type	Validation	Maps to Database Field
Full Name	Text field	Required, max 100 characters	USER_ACCOUNT.full_name
Email	Text field	Required, valid email format, must be unique	USER_ACCOUNT.username
Password	Password field	Required, min 6 characters	USER_ACCOUNT.password_hash
Sign Up Button	Submit button	Creates new user account	–

Referenced in GUI: *See Figma design – Sign-Up Page*

3.3 Sell / Product Submission Page (Input Interaction #3)



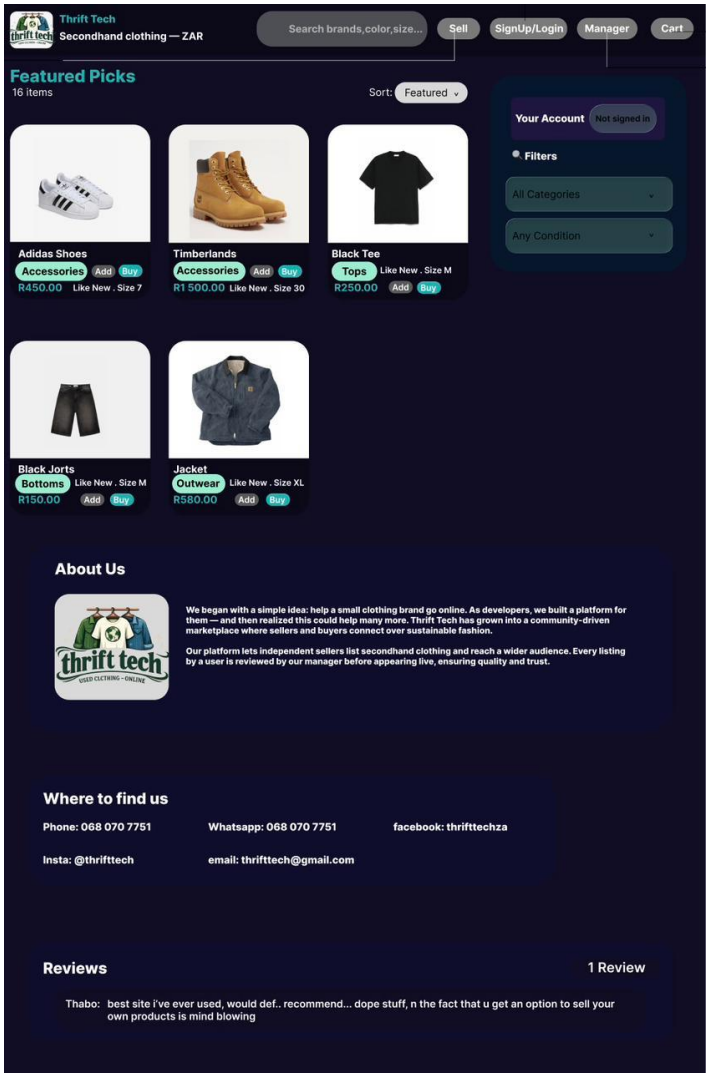
Screen Name: Sell / Product Submission Page
Purpose: Enables users to list their own second-hand items. Collects all item details and submits the listing to the manager for approval before it appears on the main site.

Control	Type	Validation	Maps to Database Field
Price	Number field	Required, ≥ 0, 2 decimal places	INVENTORY_ITEM.price
Category	Dropdown	Required (Tops, Bottoms, Footwear, Outerwear, Accessories)	INVENTORY_ITEM.category

Control	Type	Validation	Maps to Database Field
Condition	Dropdown	Required (Excellent, Good, Fair, Poor)	INVENTORY_ITEM.condition_grade
Size	Text field	Required	INVENTORY_ITEM.size_label
Image	File upload	Optional, image file type	(Stored separately as file path)
Description	Text area	Optional, max 500 characters	INVENTORY_ITEM.item_name / description
Submit Button	Submit button	Creates pending item record for manager approval	INVENTORY_ITEM.is_active = FALSE (pending)

Referenced in GUI: *See Figma design – Sell / Product Submission Page*

3.4 Home / Product Listing Page (Input Interaction #4)

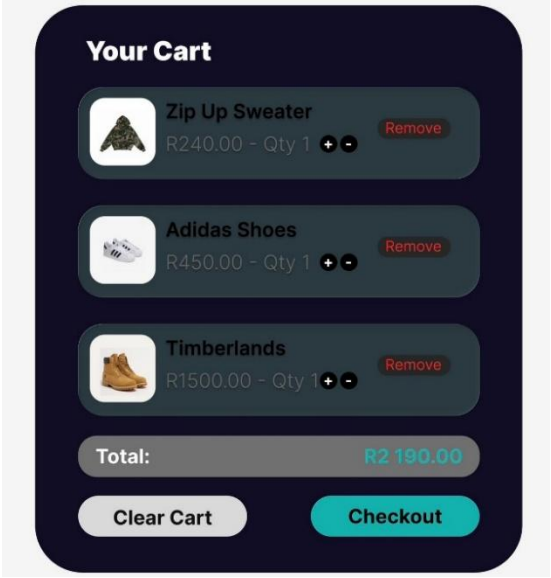

Thrift Tech
Secondhand clothing — ZAR
Search brands,color,size...
Sell
SignUp/Login
Manager
Cart
Featured Picks
16 items
Sort: Featured
Your Account
Not signed in
Filters
All Categories
Any Condition
Adidas Shoes
Accessories Add Buy
R450.00 Like New . Size 7
Timberlands
Accessories Add Buy
R1 500.00 Like New . Size 30
Black Tee
Tops Like New . Size M
R250.00 Add Buy
Black Jorts
Bottoms Like New . Size M
R150.00 Add Buy
Jacket
Outwear Like New . Size XL
R580.00 Add Buy
About Us
thrift tech
USED CLOTHING - ONLINE
We began with a simple idea: help a small clothing brand go online. As developers, we built a platform for them — and then realized this could help many more. Thrift Tech has grown into a community-driven marketplace where sellers and buyers connect over sustainable fashion.
Our platform lets independent sellers list secondhand clothing and reach a wider audience. Every listing by a user is reviewed by our manager before appearing live, ensuring quality and trust.
Where to find us
Phone: 068 070 7751
Whatsapp: 068 070 7751
facebook: thrifttechza
Insta: @thrifttech
email: thrifttech@gmail.com
Reviews
1 Review
Thabo: best site i've ever used, would def.. recommend... dope stuff, n the fact that u get an option to sell your own products is mind blowing

Screen Name: Home / Product Listing Page
Purpose: Serves as the public storefront. Showcases featured products with prices, sizes, and conditions. Users can browse, search, or click on products to view details or add to cart.

Control	Type	Validation	Maps to Database Field
Search Bar	Text field	Optional, free text	INVENTORY_ITEM.item_name
Category Filter	Dropdown / Buttons	Optional	INVENTORY_ITEM.category
Add to Cart Button	Button (per product)	Requires item in stock	INVENTORY_ITEM.item_id, quantity
Product Click	Link/Button	Navigates to product detail	INVENTORY_ITEM.item_id

Referenced in GUI: *See Figma design – Home / Product Listing Page*

3.5 Shopping Cart Page (Input Interaction #5)



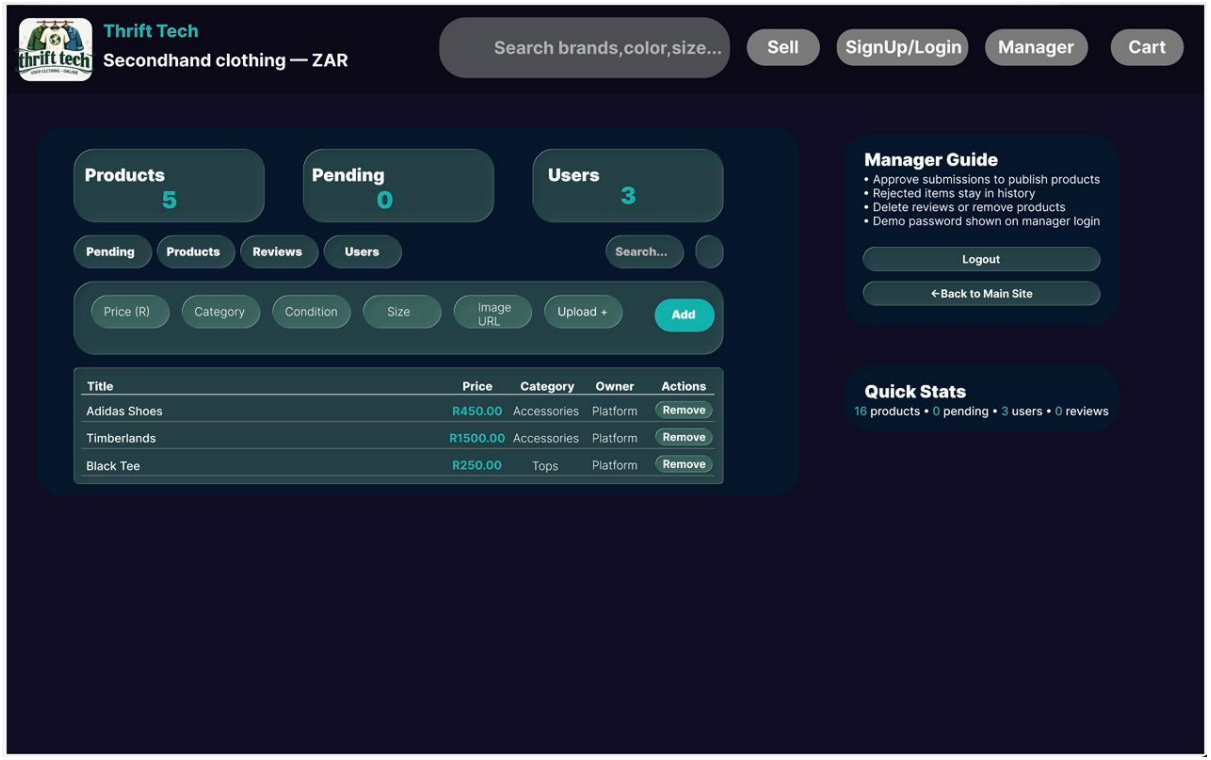
Screen Name: Shopping Cart Page
Purpose: Allows users to review items they intend to purchase. Displays selected products, quantities, individual prices, and total cost, with options to remove items or proceed to checkout.

Control	**Type**	**Validation**	**Maps to Database Field**
Quantity Selector	Number field (dropdown or +/- buttons)	≥ 1, ≤ available stock	SALE_ITEM.quantity_sold
Remove Item Button	Button	Removes item from cart	–

Control	Type	Validation	Maps to Database Field
Update Cart Button	Button	Updates quantities	SALE_ITEM.quantity_sold
Proceed to Checkout Button	Button	Creates sale transaction	SALE_TRANSACTION, SALE_ITEM
Payment Method Selector	Dropdown	Required (Cash/Card/EFT)	SALE_TRANSACTION.payment_method

Referenced in GUI: *See Figma design – Shopping Cart Page*

3.6 Admin / Manager Dashboard (Input Interaction #6)



Screen Name: Admin / Manager Dashboard
Purpose: Acts as the backend control panel for the platform manager. Allows the manager to approve or reject pending product submissions, view quick stats (products, users), and manually remove existing products or reviews before they go live.

Control	Type	Validation	Maps to Database Field
Approve Button	Button (per pending item)	Changes item status to approved	INVENTORY_ITEM.is_active = TRUE

Control	Type	Validation	Maps to Database Field
Reject Button	Button (per pending item)	Deletes or marks as rejected	INVENTORY_ITEM.is_active = FALSE (or delete)
Remove Product Button	Button (per live product)	Requires confirmation, soft delete	INVENTORY_ITEM.is_active = FALSE
Remove Review Button	Button (per review)	Deletes customer review	(Reviews table – future enhancement)
Stats Display	Labels/Text (read-only)	Displays counts: products, users, pending approvals	SELECT COUNT(*) queries

Referenced in GUI: *See Figma design – Admin / Manager Dashboard*

3.7 Summary of All Input Screens

Screen	Primary Inputs	Purpose
Login Page	Full Name, Email, Password	Authenticate existing users
Sign-Up Page	Full Name, Email, Password	Register new users

Screen	Primary Inputs	Purpose
Sell / Product Submission Page	Price, Category, Condition, Size, Image, Description	List new second-hand items (pending approval)
Home / Product Listing Page	Search, Category Filter, Add to Cart	Browse and select products
Shopping Cart Page	Quantity Selector, Remove Item, Payment Method	Review and purchase items
Admin / Manager Dashboard	Approve, Reject, Remove Product, Remove Review	Manage pending items and content

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4. User Interaction Design – Request Interactions (Option 2: Complete GUI Definition)

Request interactions represent service requests to the system, including functional processing (calculating totals, generating reports) and outputs (screen displays, printed reports). The following sections define all request interactions mapped to the GUI.

4.1 Login Request (Request Interaction #1)

Trigger: User clicks "Login" button on Login Page
Processing: System validates credentials against USER_ACCOUNT table; creates session with role

Output: Redirects to Dashboard (Admin Dashboard for managers, Home Page for regular staff/customers)
Error Output: "Invalid email or password" message displayed on Login Page

4.2 Sign-Up Request (Request Interaction #2)

Trigger: User clicks "Sign Up" button on Sign-Up Page
Processing: System checks if email already exists; creates new USER_ACCOUNT record with role = "User"
Output: Redirects to Login Page with success message "Account created. Please log in."
Error Output: "Email already registered" or "Please fill in all fields"

4.3 Product Submission Request (Request Interaction #3)

Trigger: User clicks "Submit" button on Sell / Product Submission Page
Processing: System creates new INVENTORY_ITEM record with is_active = FALSE (pending approval); stores image file reference
Output: "Your item has been submitted for approval. You will be notified when it is live."
Error Output: Validation errors for missing required fields

4.4 Search Inventory Request (Request Interaction #4)

Trigger: User enters text in Search Bar and clicks search icon (or presses Enter) on Home Page
Processing: System queries INVENTORY_ITEM table WHERE item_name LIKE '%keyword%' AND is_active = TRUE
Output: Filtered product listing showing matching items with name, price, size, condition
Performance Requirement: Must return results in ≤10 seconds (FR-02)

4.5 Add to Cart Request (Request Interaction #5)

Trigger: User clicks "Add to Cart" button on a product on Home Page
Processing: System adds item to shopping cart session; checks stock availability
Output: Confirmation message "Item added to cart"; cart icon updates with item count
Error Output: "Item out of stock" if quantity = 0

4.6 Update Cart / Remove Item Request (Request Interaction #6)

Trigger: User changes quantity or clicks "Remove Item" on Shopping Cart Page
Processing: System updates cart session; recalculates subtotal and total
Output: Updated cart display with new quantities and total amount; item removed from list if Remove clicked

4.7 Checkout / Process Sale Request (Request Interaction #7)

Trigger: User clicks "Proceed to Checkout" button on Shopping Cart Page
Processing: System creates SALE_TRANSACTION record; creates SALE_ITEM records for each cart item; decrements INVENTORY_ITEM.quantity for each item sold; generates receipt_number; clears cart session
Output: On-screen receipt showing itemised list, total, payment method, date, receipt number; automatic print dialog
Error Output: "Insufficient stock for [item name]" if stock changed between add-to-cart and checkout

4.8 Approve / Reject Product Request (Request Interaction #8)

Trigger: Manager clicks "Approve" or "Reject" button on Admin Dashboard
Processing: If Approve: updates INVENTORY_ITEM.is_active = TRUE; If Reject: deletes or sets is_active = FALSE
Output: Dashboard refreshes; item removed from pending list

4.9 Remove Product Request (Request Interaction #9)

Trigger: Manager clicks "Remove Product" button on Admin Dashboard
Processing: System soft-deletes item (INVENTORY_ITEM.is_active = FALSE)
Output: Product removed from live store; confirmation message; dashboard stats update

4.10 Generate Sales Report Request (Request Interaction #10)

Trigger: Manager clicks "Generate Report" button on Admin Dashboard (report section)
Processing: System aggregates data from SALE_TRANSACTION and SALE_ITEM tables for selected date range; calculates total revenue,

top categories, units sold
Output: On-screen HTML table with summary statistics and daily breakdown; PDF download button available

4.11 View Low Stock Request (Request Interaction #11)

Trigger: Manager dashboard loads (automated) OR manager clicks "Low Stock Report"
Processing: System compares each INVENTORY_ITEM.quantity with STOCK_THRESHOLD.min_quantity for its category; collects items where quantity < threshold
Output: Colour-coded alert panel on dashboard showing: item name, category, current quantity, threshold, "Order stock" suggestion

4.12 Logout Request (Request Interaction #12)

Trigger: User clicks "Logout" button (top right on Dashboard)
Processing: System destroys user session; clears session variables
Output: Redirects to Login Page

4.13 Summary of All Request Interactions

Request	**Trigger Screen**	**Processing**	**Output**
Login	Login Page	Validate credentials	Dashboard/Home redirect
Sign Up	Sign-Up Page	Create user account	Login redirect
Product Submission	Sell Page	Create pending item	Confirmation message

Request	Trigger Screen	Processing	Output
Search Inventory	Home Page (search bar)	Query INVENTORY_ITEM	Filtered product list
Add to Cart	Home Page (Add button)	Add to session	Cart updated
Update/Remove Cart	Cart Page	Recalculate totals	Updated cart display
Checkout (Process Sale)	Cart Page → Checkout	Create sale, update stock	On-screen receipt + print
Approve/Reject Product	Admin Dashboard	Update is_active flag	Dashboard refreshes
Remove Product	Admin Dashboard	Soft delete item	Product removed from store
Generate Sales Report	Admin Dashboard	Aggregate sales data	On-screen report + PDF
View Low Stock	Admin Dashboard	Compare quantity vs threshold	Alert panel
Logout	Dashboard	Destroy session	Login redirect

4.14 Request Interactions Not in Analysis Document (Added Here)

The following request interactions were not fully detailed in the Requirement Analysis document and have been added to ensure completeness:

1. Sign-Up Request
2. Update Cart / Remove Item Request
3. Approve/Reject Product Request
4. Remove Product Request
5. Logout Request

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5. Database Design – Database Tables

The database schema for TTISS consists of **five normalised tables** in Third Normal Form (3NF). Entities were identified from the class diagram and GUI inputs (Sell Page, Cart Page, Login/Signup Pages, Admin Dashboard). Each table includes primary keys, foreign keys where applicable, and **at least three fictitious but realistic example values**.

Database Table #1: INVENTORY_ITEM

Schema:

Role	**Column Name**	**Type**	**Description**
Primary Key	item_id	INT	Unique identifier

Role	Column Name	Type	Description
Data Field	item_name	VARCHAR(100)	Clothing description / title
Data Field	category	ENUM	Tops/Bottoms/Dresses/Outerwear/Footwear/Accessories
Data Field	size_label	VARCHAR(10)	Size (S, M, L, 32, EU42, etc.)
Data Field	condition_grade	ENUM	Excellent/Good/Fair/Poor
Data Field	price	DECIMAL(10,2)	Selling price in ZAR
Data Field	quantity	INT	Current stock count
Data Field	image_path	VARCHAR(255)	Optional file path to uploaded image
Data Field	description	TEXT	Optional item description (max 500 chars)
Data Field	date_added	DATETIME	When item was added
Data Field	is_active	BOOLEAN	TRUE = live on Home Page, FALSE = pending or removed

Role	Column Name	Type	Description
Foreign Key	submitted_by	INT (FK → USER_ACCOUNT)	User who submitted the item

Example Data (three fictitious rows):

item_id	item_name	category	size_label	condition_grade	price	quantity	image_path	description	date_added	is_active	submitted_by
101	Levi's 501 Jeans	Bottoms	32	Good	249.99	5	/images/101.jpg	Classic blue jeans	2026-04-15 10:30:00	TRUE	2
102	Vintage Nike Hoodie	Outerwear	L	Excellent	399.00	2	/images/102.jpg	Rare 90s hoodie	2026-04-10 14:20:00	TRUE	2
103	Zara Blazer	Tops	M	Fair	150.00	0	/images/103.jpg	Office wear	2026-03-28	TRUE	3

item_id	item_name	category	size_label	condition_grade	price	quantity	image_path	description	date_added	is_active	submitted_by
									09:15:00		

Database Table #2: SALE_TRANSACTION

Schema:

Role	Column Name	Type	Description
Primary Key	sale_id	INT	Unique transaction ID
Foreign Key	user_id	INT (FK → USER_ACCOUNT)	Customer who made the purchase
Data Field	sale_date	DATETIME	Date and time of transaction
Data Field	total_amount	DECIMAL(10,2)	Total sale value

Role	Column Name	Type	Description
Data Field	payment_method	ENUM	Cash/Card/EFT
Data Field	receipt_number	VARCHAR(20)	Human-readable receipt reference (UNIQUE)

Example Data (three fictitious rows):

sale_id	user_id	sale_date	total_amount	payment_method	receipt_number
5001	2	2026-05-09 11:45:00	649.99	Card	RCP-20260509-001
5002	2	2026-05-09 14:20:00	150.00	Cash	RCP-20260509-002
5003	3	2026-05-08 16:10:00	399.00	EFT	RCP-20260508-023

Database Table #3: SALE_ITEM

Schema:

Role	Column Name	Type	Description
Primary Key	sale_item_id	INT	Unique line item ID
Foreign Key	sale_id	INT (FK → SALE_TRANSACTION)	Links to parent sale
Foreign Key	item_id	INT (FK → INVENTORY_ITEM)	Links to product sold
Data Field	quantity_sold	INT	Number of units
Data Field	unit_price	DECIMAL(10,2)	Price at time of sale (snapshot)
Generated	line_total	DECIMAL(10,2)	quantity_sold * unit_price (stored)

Example Data (three fictitious rows):

sale_item_id	sale_id	item_id	quantity_sold	unit_price	line_total
10001	5001	101	1	249.99	249.99
10002	5001	102	1	399.00	399.00

sale_item_id	sale_id	item_id	quantity_sold	unit_price	line_total
10003	5002	103	1	150.00	150.00

Database Table #4: USER_ACCOUNT

Schema:

Role	Column Name	Type	Description
Primary Key	user_id	INT	Unique user ID
Unique Key	username (email)	VARCHAR(50)	Login email (unique)
Data Field	password_hash	VARCHAR(255)	bcrypt hash of password
Data Field	full_name	VARCHAR(100)	Display name
Data Field	role	ENUM	User/Staff/Manager/Admin
Data Field	is_active	BOOLEAN	Deactivated accounts cannot log in

Role	Column Name	Type	Description
Data Field	created_at	DATETIME	Account creation date

Example Data (three fictitious rows):

user_id	username	password_hash	full_name	role	is_active	created_at
1	admin@thrift.com	$2y10$abc...	System Admin	Admin	TRUE	2026-03-01 09:00:00
2	john@example.com	$2y10$def...	John Dlamini	User	TRUE	2026-03-15 10:30:00
3	mary@thrift.com	$2y10$ghi...	Mary Molefe	Manager	TRUE	2026-04-01 14:00:00

Database Table #5: STOCK_THRESHOLD

Schema:

Role	Column Name	Type	Description
Primary Key	threshold_id	INT	Unique threshold ID

Role	Column Name	Type	Description
Data Field	category	ENUM	Must match INVENTORY_ITEM.category
Data Field	min_quantity	INT	Alert when quantity drops below this (default 5)
Foreign Key	updated_by	INT (FK → USER_ACCOUNT)	Manager who last changed this threshold
Data Field	updated_at	DATETIME	Timestamp of last change

Example Data (three fictitious rows):

threshold_id	category	min_quantity	updated_by	updated_at
1	Tops	3	3	2026-04-10 09:00:00
2	Bottoms	3	3	2026-04-10 09:00:00
3	Outerwear	2	3	2026-04-15 11:30:00

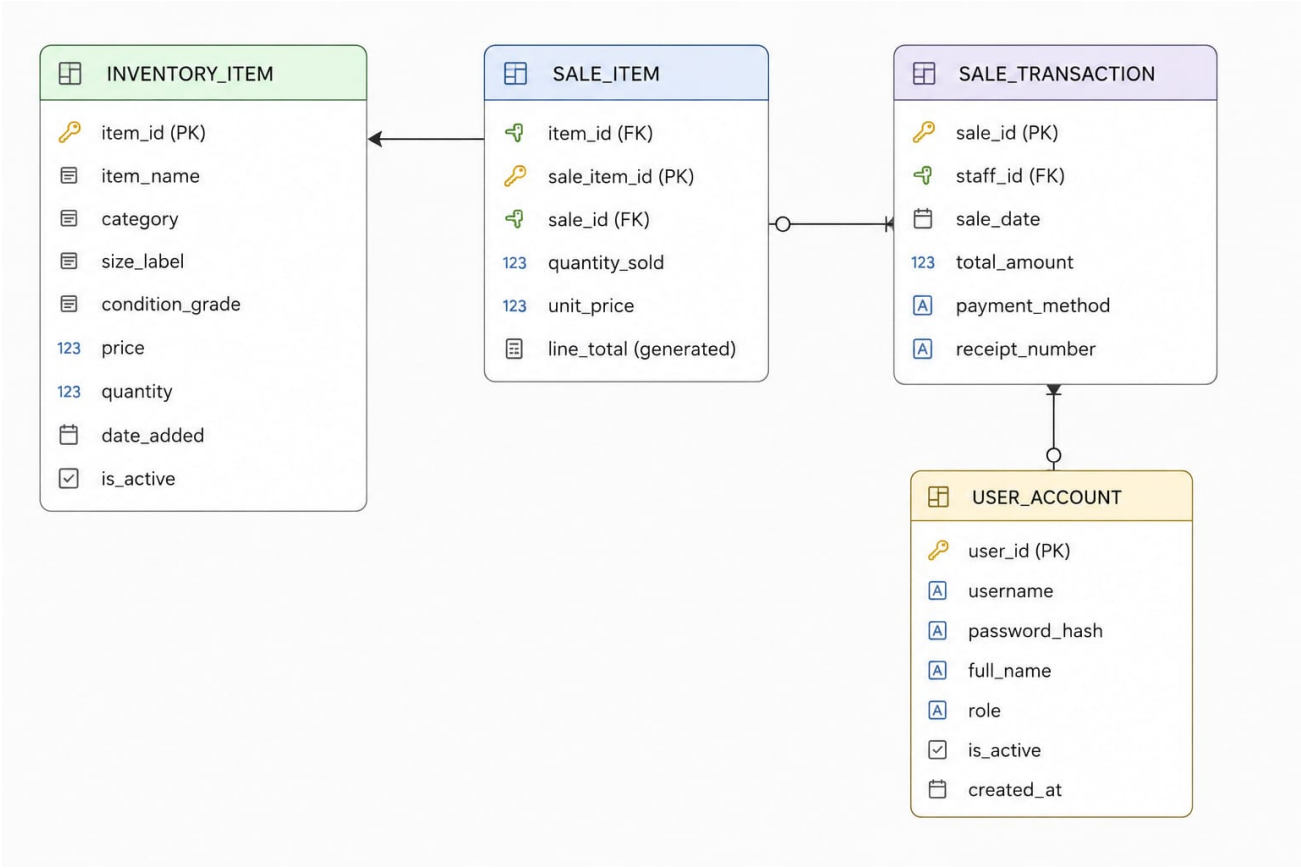
Note: Other categories (Dresses, Footwear, Accessories) use the default value of 5 if not explicitly set.

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6. Database Design – ERD Design

The Entity Relationship Diagram (ERD) shows the relationships between the five database tables. Cardinality and foreign key constraints are clearly represented.

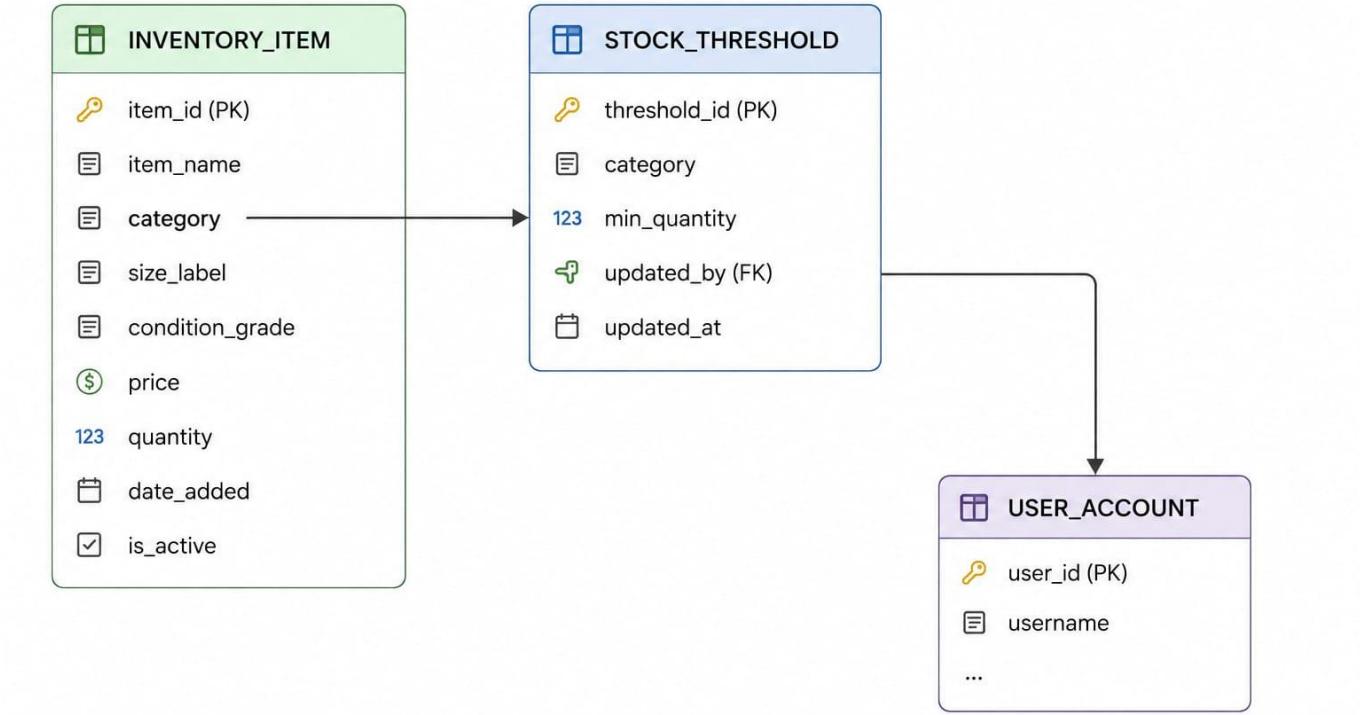
ERD Diagram #1: Core Sales and Inventory Relationships



Relationships:

- **INVENTORY_ITEM** (1) → **SALE_ITEM** (Many)
- **SALE_TRANSACTION** (1) → **SALE_ITEM** (Many)
- **USER_ACCOUNT** (1) → **SALE_TRANSACTION** (Many)
- **USER_ACCOUNT** (1) → **INVENTORY_ITEM** (Many)

ERD Diagram #2: Stock Threshold and Inventory Category Relationship



Relationships:

- **INVENTORY_ITEM.category** → **STOCK_THRESHOLD.category** (Many-to-One)
- **USER_ACCOUNT** (1) → **STOCK_THRESHOLD** (Many)

Relationships Summary Table:

Entity A	Relationship	Entity B	Cardinality
SALE_TRANSACTION	contains	SALE_ITEM	1 : Many
INVENTORY_ITEM	appears in	SALE_ITEM	1 : Many
USER_ACCOUNT	makes	SALE_TRANSACTION	1 : Many
USER_ACCOUNT	submits	INVENTORY_ITEM	1 : Many
USER_ACCOUNT INVENTORY_ITEM.category	configures governed by	STOCK_THRESHOLD STOCK_THRESHOLD.category	1 : Many Many : 1

Normalisation verification: All tables are in 3NF. No repeating groups, no partial dependencies, no transitive dependencies.

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7. Report Design

The TTISS generates three types of reports for the client (Thrift Tech management). Each report is designed to be accurate, usable, and meet the requirements specified in the Project Plan and Requirement Analysis Document.

Report 1: Weekly Sales Report

Client need: Management requires visibility into revenue, transaction volumes, and top-selling categories (FR-04).

Data sources: SALE_TRANSACTION, SALE_ITEM, INVENTORY_ITEM

Report Layout:

THRIFT TECH – WEEKLY SALES REPORT
Date range: [Start Date] to [End Date]
Generated: [Current Date]
Summary Statistics
Total revenue: R 12,450.00
Total transactions: 48
Average transaction: R 259.38
Total units sold: 142

Top 3 Categories (by units sold)
1. Bottoms - 62 units
2. Tops - 41 units
3. Outerwear - 24 units
Daily Breakdown
2026-05-04: 12 transactions, R3,200, 35 units 2026-05-05: 10 transactions, R2,800, 30 units
2026-05-06: 9 transactions, R2,450, 28 units

Presentation format: On-screen HTML table + PDF download + Print

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Report 2: Low-Stock Alert Report

Client need: Staff and managers must be notified when inventory falls below threshold (FR-05).

Data sources: INVENTORY_ITEM, STOCK_THRESHOLD

Report Layout (dashboard alert panel):

Low Stock Alerts (Updated: [Current Time])
Bottoms (Threshold: 3): Lee Cooper Chinos - Stock: 2 → BELOW THRESHOLD
Outerwear (Threshold: 2): North Face Puffer - Stock: 1 → BELOW THRESHOLD
Tops (Threshold: 3): Zara Blazer - Stock: 0 → OUT OF STOCK
Suggested order quantities:
Lee Cooper Chinos: +5 units
North Face Puffer: +3 units
Zara Blazer: +5 units

Presentation format: Colour-coded panel on Admin Dashboard + full page view

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Report 3: Sale Receipt

Client need: Customers require proof of purchase (FR-03).

Data sources: SALE_TRANSACTION, SALE_ITEM, INVENTORY_ITEM, USER_ACCOUNT

Receipt Layout:

THRIFT TECH – BRAAMFONTEIN
Receipt #: RCP-20260509-001
Date: 2026-05-09 11:45:00
Customer: John Dlamini

Item	QTY	Price	Total
Levi's 501 Jeans	1	249.99	249.99
Vintage Nike Hoodie	1	399.00	399.00
TOTAL			**R 649.99**

| Payment method: Card | Amount paid: R650.00 | Change: R0.01 |

Presentation format: On-screen receipt modal + automatic print dialog

Report Requirements Summary

Report	Data Sources	Format	Delivery Method	Meets Client Need
Weekly Sales Report	SALE_TRANSACTION, SALE_ITEM, INVENTORY_ITEM	HTML + PDF	Admin Dashboard	Yes
Low-Stock Alert	INVENTORY_ITEM, STOCK_THRESHOLD	Colour-coded panel	Admin Dashboard	Yes
Sale Receipt	All four main tables	Modal + print	Post-checkout	Yes

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8. Document Formatting / Layout

This document follows a professional layout with clear structure as specified in Section 7 of the WIL Requirements.

Requirement	Status	Evidence
Cover page with team number, team name, member names and student numbers, client name, system logo/name	✓	Cover page at beginning

Requirement	Status	Evidence
Numbered index of contents	✓	Table of Contents with sections 1 through 8
Introduction directed at company management	✓	Section 1 addressed to Thrift Tech management
Logical architectural design – high and low level	✓	Sections 2.1 and 2.2
User interaction design – input interactions (Option 2)	✓	Section 3 – six GUI screens
User interaction design – request interactions (Option 2)	✓	Section 4 – twelve request interactions
Database design – tables (minimum 4 tables, 3NF, example data)	✓	Section 5 – five tables with schema and 3 example rows each
Database design – ERD design	✓	Section 6 – two ERD diagrams and relationship summary
Report design	✓	Section 7 – three reports

Requirement	Status	Evidence
Consistent headings and numbering	✓	All sections use hierarchical numbering
Tables and diagrams clear	✓	All tables use consistent format
References list	✓	Below
Professional overall layout	✓	Cover page, table of contents, clear section breaks

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References:

Thrift Tech Project Plan Document (XISD TASK 1_260509_211004.pdf). 2026.

Thrift Tech Requirement Analysis Document (XISD TASK2_260509_210939.pdf). 2026.

Figma Prototype – Thrift Tech Store. 2026. Available at: https://www.figma.com/design/vq5z23Yxmm94fTEwkLHr6q/thrift_tech_store?node-id=0-1&t=JeE2JPcH6ouuNxe6-1

Appendix A – Technology Stack Summary

Layer	Technology	Justification
Frontend	HTML5, CSS3, JavaScript	Runs in any browser on existing hardware
Backend	PHP	Lightweight server-side processing
Database	MySQL	Free, well-documented, industry-standard
Prototyping	Figma	Free browser-based UI prototyping
Version Control	GitHub	Free, supports team collaboration

End of System Design Document